

Adaptive Prompts for Personalized Learning

Objective:

To create a dynamic and adaptive prompt system tailored for High School Algebra—specifically solving single-variable linear equations. The system aims to personalize the learning experience by adapting prompts in real-time based on the learner's responses, proficiency level, preferences, and learning progress.

Task Overview:

The task involves designing a set of adaptive prompts that cater to learners' individual needs. The prompts dynamically change depending on how the learner interacts with them, adjusting in complexity and support based on performance and preferences.

Learning Domain:

High School Algebra – Solving Linear Equations

Prompt Design Strategy:

The adaptive prompt design considers multiple factors: learner proficiency (novice, intermediate, advanced), response accuracy, learning preferences (visual, text-heavy, step-by-step, challenge-based), and progress over time. Based on these inputs, the system dynamically selects the most appropriate prompt path.

Example of Adaptive Prompt Flow:

- Initial Diagnostic Prompt (Skill Check):

Hi! Let's find out your current skill level. Solve this equation:

$$2x + 3 = 11$$

Show your steps.

- Adaptation Based on Response:

Correct (Intermediate Proficiency):

Nice work! Let's try one that's slightly more complex:

$$3(x - 2) = 2x + 4$$

Remember to distribute first.

Incorrect (Novice Proficiency):

No problem! Let's go over how to isolate x step-by-step.

Step 1: What's $11 - 3$?

Step 2: Now divide the result by 2.

Visual Learner Option:

Here's a balance scale showing both sides of the equation.

Can you figure out what we need to subtract to keep the scale balanced?

Prompt Adaptation Algorithm (Simplified):

```
def adapt_prompt(response_correct, learner_level, preferences):  
    if not response_correct:  
        if learner_level == "novice":  
            return "Break down problem into smaller steps"  
        else:  
            return "Offer hint or simplified problem"  
  
    else:  
        if learner_level == "advanced":  
            return "Increase difficulty"  
        else:  
            return "Same difficulty with varied format"  
  
    if preferences == "visual":  
        return "Add diagram or visual aid"
```

Tracking Learner Progress:

The system tracks key metrics such as:

- Accuracy over time
- Preferred learning format (visual, step-by-step)
- Common error patterns
- Time taken per question

Evaluation and Reflection:

This adaptive prompt system demonstrates the potential of AI to support personalized education. It adjusts in real time to address individual learning needs, scaffold learning experiences, and keep learners engaged through appropriate challenges. Further development into a functional prototype would enhance usability and accessibility, making the system intuitive and more impactful.

Submission:

Please provide a shared link to the interactive prompt system (e.g., GPT Playground, Google Colab, or another interface) that demonstrates the adaptive logic in action. Include at least three distinct branching paths based on user interaction.